Lumi Ride – Booking & Service Management System Specification

Version 1.2 (Draft)

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1. Introduction

This document outlines the current operational process, limitations, and proposed functional system requirements for Lumi Ride's booking, scheduling, customer communications, and CRM infrastructure. The goal is to transition from a largely manual operational workflow into an integrated, automated, customer-centric service booking platform, aligned with contemporary best-practice models used in transport, disability services, logistics, and on-demand service industries.

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2. Current State Analysis (Existing System)

2.1 Overview

Lumi Ride currently operates a fully manual end-to-end workflow for:

- Booking creation
- Customer updates
- Driver communication
- Scheduling & shift allocation
- Data entry
- CRM functions

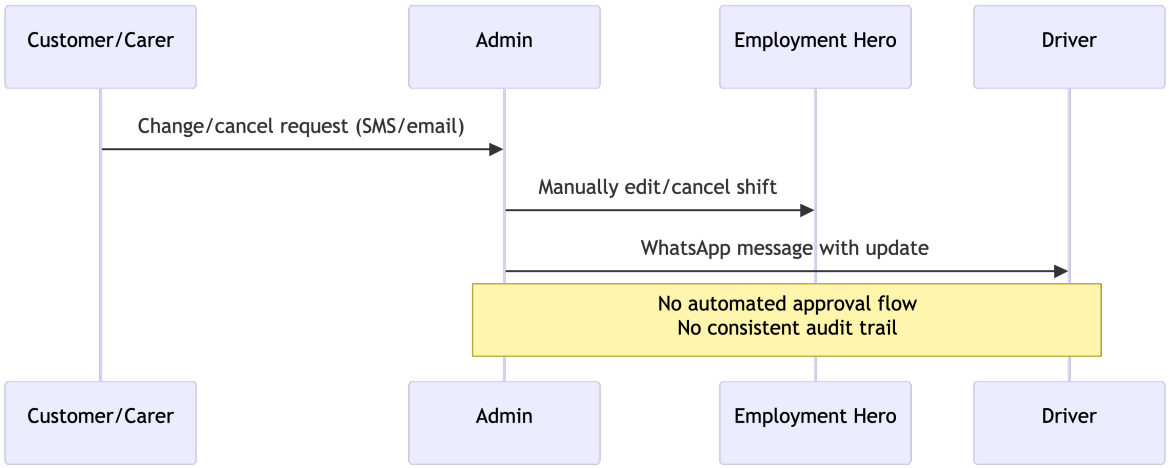
All administration is performed manually using:

- Employment Hero (EH) — for shift creation, roster management, and driver assignments
- WhatsApp & SMS (via Twilio) — for driver notifications
- Email & SMS — for customer communication
- Manual spreadsheets or templates — for job details

2.2 Current Booking & Scheduling Workflow

1. Customer creates or updates recurring bookings
    - Usually via SMS, email, or phone call
    - Admin staff manually collect details
2. Roster creation
    - Recurring weekly schedules created in Employment Hero
    - Distributed to drivers for acceptance
3. Change requests or cancellations

    - Received via mobile text or email to info@lumiride.com
    - Admin staff manually update Employment Hero
    - Admin sends updated shift information to drivers via WhatsApp manually
4. New client onboarding
    - Triggered by a call or email
    - Admin prepares quote
    - Funding approval awaited
    - Trip details reconfirmed verbally
    - Admin manually creates customer + ride templates in Employment Hero
5. Shift → Job conversion
    - Admin manually enters each ride into the admin portal
    - Current portal is non-intuitive and inconsistent (field jumps, UI issues)
6. Driver notification
    - Triggered via automated SMS (Twilio), although email notifications may not be functioning
    - Weekly ride allocation is typically delivered the night before to reduce driver forgetfulness
7. Customer communication
    - Almost entirely unstructured and not automatically logged
    - No written audit trail beyond email or SMS history



3. Pain Points

Operational

- Excessive manual data entry
- Duplication across Employment Hero, templates, and internal systems
- UI inconsistencies in admin portal slow down processing
- Weak audit & traceability for bookings and changes
- No versioning or change history for bookings

Customer-facing

- Customers cannot view their own schedules
- No self-service options (book, cancel, modify)
- High reliance on verbal communication introduces risk
- Lack of legal documentation trail

Driver-facing

- No weekly manifest or consolidated job view
- Must rely on SMS or Employment Hero login
- Communication channels fragmented

Strategic

- No CRM system
- Contact information scattered across various systems
- No central store for:
    - riders
    - carers
    - plan managers
    - NDIS contact points
    - accounting parties
- No automation around contact creation, updates, or exports

4. Future System Vision

Lumi Ride requires a modern, integrated service booking platform that covers:

- Automated booking management
- Self-service customer portal
- Driver portal/manifest
- Unified CRM
- Integration with Employment Hero or replacement scheduling API
- Automated change detection + approval workflow
- Automated communication via SMS, push notification, or email

This aligns with best-practice models used by:

- rideshare platforms (Uber, Ola)
- disability transport management systems
- logistics fleet scheduling tools
- healthcare appointment systems

5. Proposed Projects

Project #1 – Automation & Portal Development

Objectives

- Reduce reliance on manual data entry
- Provide structured, auditable communication
- Improve operational efficiency

- Deliver customer and driver self-service capabilities

Key Functions Needed

1. Employment Hero API integration
    - Extract & sync shift/ride data
    - Push job updates
    - Generate manifests for drivers (Google Sheets or integrated web app)
2. Driver portal / manifest system
    - Weekly and daily schedule views
    - Accept/decline functionality
    - Real-time updates
3. Customer portal
    - View all upcoming rides
    - Make changes or cancellations
    - Submit new ride requests
    - Secure messaging
    - Notifications
    - Must include approval workflow so changes don’t disrupt operations

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Project #2 – CRM System

Key Objectives

- Centralise all customer-related information
- Capture all entities involved in a rider’s care & funding
- Streamline communication
- Support compliance and reporting

Core CRM Requirements

- Contact management:
    - Rider
    - Carer
    - Plan manager
    - NDIS stakeholders
    - Accounting contacts
- Email, phone, address, role, relationship metadata
- Auto-creation of contacts when entering new ride bookings
- Timestamps for creation & modification
- Searchable notes per client or ride
- Export to CSV for bulk email or reporting
- Integration with booking system for unified workflow

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6. Functional Requirements (FR)

6.1 Customer Portal

- Secure login (email/SMS-based authentication)
- View upcoming & historical rides
- Request modifications or cancellations
- Create new booking requests
- Receive automated confirmations
- Upload documentation (optional)
- View account profile & NDIS-related information

6.2 Driver Portal

- Mobile-friendly real-time manifest
- One-tap acknowledgement of shifts
- Notifications for changes
- Route information (optional future phase)

6.3 Admin Portal Enhancements

- Clean, consistent UI for job entry
- Automated data validation
- Integrated scheduling overview
- Change approval workflow
- Full change history (audit trail)
- Trigger automated communication events

6.4 Communication System

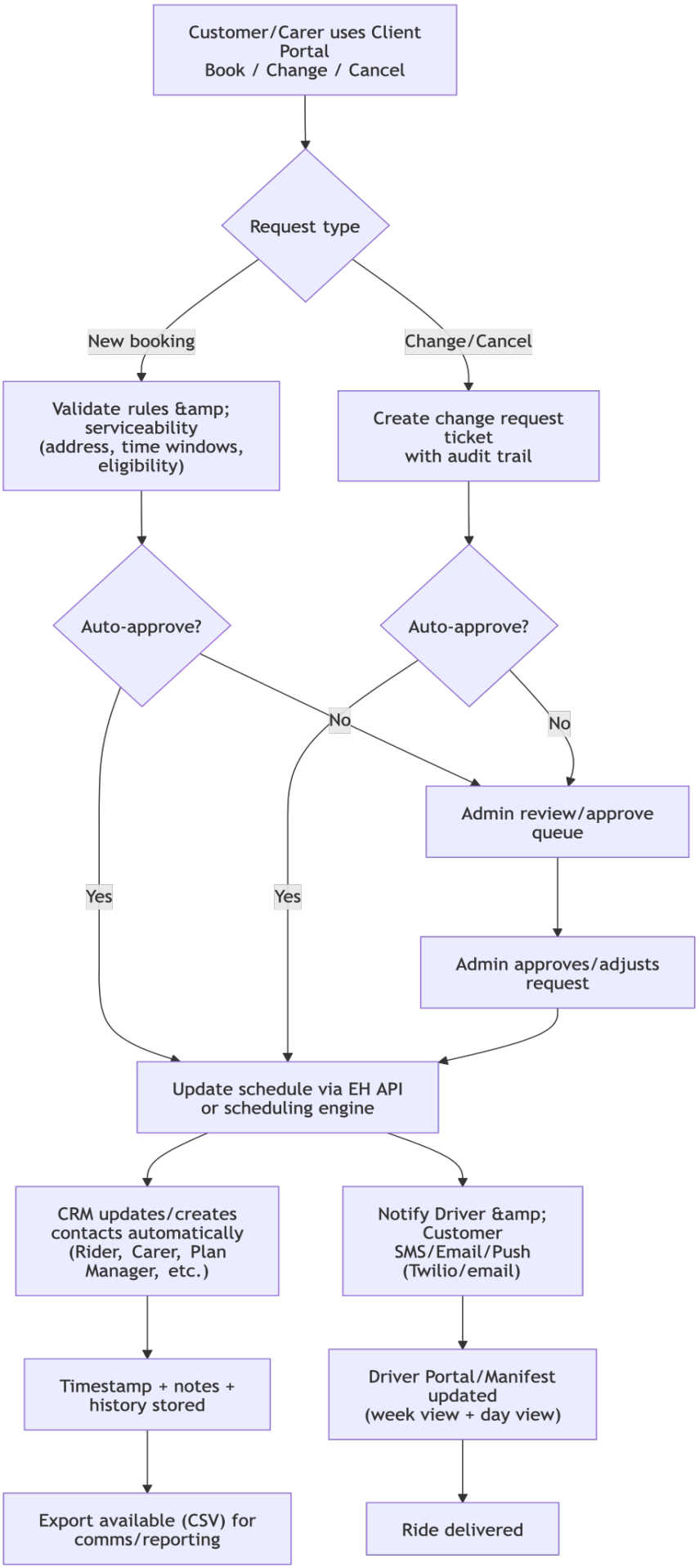
- SMS + email + WhatsApp integrations
- All messages logged in CRM
- Automatic notifications for:
    - new bookings
    - modifications
    - cancellations
    - reminders (driver & customer)

6.5 CRM

- Central records of all contacts
- Role-based tagging
- Notes + timestamping
- Export to CSV
- Automated contact creation when entering jobs

6.6 Integration Layer

- Employment Hero API
- Twilio SMS
- Optional: Google Sheets, Gmail/Outlook, Xero


Customer/Carer uses Client Portal
Book / Change / Cancel
Request type
New booking
Change/Cancel
Validate rules & serviceability (address, time windows, eligibility)
Create change request ticket with audit trail
Auto-approve?
Auto-approve?
No
No
Yes
Yes
Admin review/approve queue
Admin approves/adjusts request
Update schedule via EH API or scheduling engine
CRM updates/creates contacts automatically (Rider, Carer, Plan Manager, etc.)
Notify Driver & Customer SMS/Email/Push (Twilio/email)
Timestamp + notes + history stored
Driver Portal/Manifest updated (week view + day view)
Export available (CSV) for comms/reporting
Ride delivered

7. Non-Functional Requirements (NFR)

- Security:
  - MFA, encrypted storage, NDIS data-handling standards
- Scalability:
  - Handle growth in riders/drivers without performance loss
- Performance:
  - Page responses under 2 seconds
- Reliability:
  - 99.9% uptime
- Auditability:
  - Every change to a booking must be logged
- Usability:
  - Simple UX for customers and carers
- Accessibility:
  - WCAG 2.1 AA compliant
- Mobile Friendly:
  - All portals must be usable on phones

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8. Best-Practice Benchmarks for Booking Systems

Modern service booking systems typically include:

✔ Customer self-service portal
✔ Real-time booking updates
✔ Automatic SMS/email/push notifications
✔ Mobile-friendly driver app
✔ Audit trail of all changes
✔ Role-based access control
✔ API integrations with payroll, scheduling, CRM
✔ Analytics dashboard
✔ Continuous syncing across platforms

These features are standard across:

- rideshare providers
- disability transport providers
- modern healthcare appointment systems
- logistics scheduling platforms

This specification moves Lumi Ride toward that benchmark.

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9. Recommended Next Steps

1. Technical Feasibility Study
   - Confirm Employment Hero API capabilities
   - Evaluate whether EH remains the scheduling source of truth
2. System Architecture Plan

    - Decide on in-house vs commercial CRM
    - Define integration layer / middleware
3. UX Wireframes
    - Customer portal
    - Driver portal
    - Admin updates
4. Implementation Roadmap
    - Phase 1: API + Driver Manifest
    - Phase 2: Customer Portal
    - Phase 3: CRM Integration
    - Phase 4: Automation & analytics
5. Budget & Vendor Assessment
    - Internal dev vs third-party platform extensions